Lecture 7

The group algebra

Objectives. *Lecture 6 showed that G has at most as many irreducible representations as conjugacy classes. The* group algebra $\mathbb{C}[G]$ *contains every irreducible representation and explains why their number equals the number of classes. It is isomorphic to the direct sum of the matrix algebras* $\operatorname{End}(U)$*, and its centre counts the classes. The matrix blocks* $\operatorname{End}(U)$ *also compute the commutant* $\operatorname{End}_G(V)$*, and decide when restriction to a subgroup is multiplicity-free, as Lecture 10 needs for S_n.*

By the end of this lecture you will be able to:

- Prove $\sum_U (\dim U)^2 = |G|$ from the regular module, and use it to prove that $\mathbb{C}[G] \cong \bigoplus_U \operatorname{End}(U)$ as algebras.
- Prove that the class sums are a basis of the centre of $\mathbb{C}[G]$, that G has as many irreducible representations as classes, and the column orthogonality relations; compute central idempotents.
- Split a representation into isotypic components, and compute its commutant $\operatorname{End}_G(V) \cong \bigoplus_U \operatorname{End}(M_U)$.
- Prove that every irreducible representation of G restricts to $H \le G$ without multiplicity exactly when $\mathbb{C}[G]^H$ is commutative, and check it for $S_2 \le S_3$.

7.1 Algebras and modules

Throughout, G is finite, and vector spaces are complex and finite-dimensional.

Definition 7.1 (Algebras)**.** An *algebra* is a vector space A with a bilinear, associative multiplication $(a, b) \mapsto ab$ that has an identity element. A *subalgebra* is a subspace closed under multiplication and containing the identity. An *algebra homomorphism* is a linear map $\varphi\colon A \to B$ with $\varphi(ab) = \varphi(a)\varphi(b)$ that sends the identity to the identity.

The endomorphism space $\operatorname{End}(V)$ is an algebra under composition, with identity $\mathbb{1}$. In a basis it is the matrix algebra $M_n(\mathbb{C})$, where $n = \dim V$. The *direct sum* $A \oplus B$ of two algebras has componentwise operations.

Definition 7.2 (The group algebra)**.** The *group algebra* $\mathbb{C}[G]$ is the vector space with basis G, whose elements are the formal sums $a = \sum_{g \in G} a_g\, g$ with $a_g \in \mathbb{C}$, and with the multiplication of G extended bilinearly:

$$\Big(\sum_g a_g\, g\Big)\Big(\sum_h b_h\, h\Big) = \sum_{g,h} a_g b_h\, gh.$$

The group algebra is associative because G is. Its identity is e, and $\dim \mathbb{C}[G] = |G|$. It is commutative exactly when G is abelian.

Example 7.3 (The group algebra of C_2)**.** Let $C_2 = \{e, t\}$, with $t^2 = e$. An element of its group algebra is a formal sum $ae + bt$ with $a, b \in \mathbb{C}$. Bilinearity gives

$$(ae + bt)(ce + dt) = (ac + bd)e + (ad + bc)t.$$

The basis elements e and t are distinct vectors, even though e also serves as the identity for multiplication.

Definition 7.4 (Modules)**.** A *module* over an algebra A is a vector space V with an algebra homomorphism $\rho\colon A \to \operatorname{End}(V)$; we write $a \cdot v = \rho(a)v$. Submodules, module homomorphisms (linear maps T with $T(a \cdot v) = a \cdot Tv$), irreducible modules and direct sums are defined as for representations (§ 5.1).

Proposition 7.5 (Representations are $\mathbb{C}[G]$-modules)**.**

(i) A representation ρ of G on V extends to exactly one algebra homomorphism $\rho\colon \mathbb{C}[G] \to \operatorname{End}(V)$, namely

$$\rho\Big(\sum_g a_g\, g\Big) = \sum_g a_g\, \rho(g).$$

Conversely, an algebra homomorphism $\mathbb{C}[G] \to \operatorname{End}(V)$ restricts on G to a representation.

(ii) The subrepresentations of V are its submodules, and the intertwiners are the module homomorphisms.

Proof. (i) The extension is linear, and multiplicative:

$$\begin{aligned}\rho(ab) &= \sum_{g,h} a_g b_h\, \rho(gh)\\ &= \sum_{g,h} a_g b_h\, \rho(g)\rho(h)\\ &= \rho(a)\rho(b),\end{aligned}$$

with $\rho(e) = \mathbb{1}$. It is the only linear extension, since G spans $\mathbb{C}[G]$. Conversely, if ρ is an algebra homomorphism, then

$$\rho(g)\rho(g^{-1}) = \rho(e) = \mathbb{1},$$

so $g \mapsto \rho(g)$ is a homomorphism into $\mathrm{GL}(V)$.

(ii) A subspace preserved by every $\rho(g)$ is preserved by their linear combinations. If $T\colon U \to V$ intertwines every $g \in G$, linearity gives $T\rho_U(a) = \rho_V(a)T$ for every $a \in \mathbb{C}[G]$; the converse follows by taking $a = g$. ∎

So representations of G are $\mathbb{C}[G]$-modules, as Lecture 5 promised. For $a \in \mathbb{C}[G]$ we write

$$\chi_V(a) = \operatorname{tr} \rho(a) = \sum_g a_g \chi_V(g).$$

7.2 The regular module and the matrix blocks

Definition 7.6 (The regular module). The *regular module* is $\mathbb{C}[G]$ under left multiplication, $a \cdot x = ax$.

Restricted to G, it is the permutation representation on G (Example 5.2). Its character χ_{reg} counts fixed points (Example 5.9). If $g \neq e$, then $gx = x$ would imply $g = e$, so left multiplication by g has no fixed point. Thus

$$\chi_{\mathrm{reg}}(e) = |G|, \qquad \chi_{\mathrm{reg}}(g) = 0 \quad (g \neq e).$$

For $G = C_2 = \{e, t\}$, left multiplication by t exchanges the basis vectors e and t. The line spanned by $e + t$ is trivial, while t acts by -1 on the line spanned by $e - t$. Thus the regular representation is the sum of the two one-dimensional representations found in Example 5.18.

Proposition 7.7 (The regular module). *Each $U \in \mathrm{Irr}(G)$ occurs in $\mathbb{C}[G]$ with multiplicity $\dim U$. Hence*

$$\chi_{\mathrm{reg}} = \sum_{U \in \mathrm{Irr}(G)} \dim U \, \chi_U,$$

and

$$\sum_{U \in \mathrm{Irr}(G)} (\dim U)^2 = |G|.$$

Proof. There are finitely many irreducibles, since their number is at most the number of conjugacy classes (Corollary 6.8). The character multiplicity formula in that corollary, and the fact that χ_{reg} vanishes off e, give

$$\begin{aligned} m_U(\mathbb{C}[G]) &= \frac{1}{|G|} \sum_{g \in G} \overline{\chi_U(g)} \, \chi_{\mathrm{reg}}(g) \\ &= \frac{1}{|G|} \dim U \cdot |G| = \dim U. \end{aligned}$$

The character expansion in Corollary 6.8 gives the formula for χ_{reg}. Evaluating it at e,

$$|G| = \chi_{\mathrm{reg}}(e) = \sum_{U} \dim U \cdot \dim U. \qquad \blacksquare$$

Example 7.8 (Guard: squares do not give divisors). The sum-of-squares count alone does not show that every irreducible degree divides $|G|$: the numbers 1 and 2 satisfy

$$1^2 + 2^2 = 5,$$

but $2 \nmid 5$. No group has exactly two irreducible representations up to isomorphism, with degrees as above: a group of order 5 is cyclic (Corollary 1.17), so its irreducibles are one-dimensional (Corollary 5.17). Lecture 9 proves $\dim U \mid |G|$ by integrality.

Theorem 7.9 (The matrix blocks). *The map*

$$\Phi \colon \mathbb{C}[G] \to \bigoplus_{U \in \mathrm{Irr}(G)} \mathrm{End}(U), \qquad \Phi(a) = \big(\rho_U(a)\big)_U,$$

is an isomorphism of algebras.

Proof. *A homomorphism.* Each ρ_U is an algebra homomorphism, by Proposition 7.5(i), and the operations on the direct sum are componentwise.

Injective. Let $\Phi(a) = 0$, and let V be a representation. By Maschke's theorem, V is a direct sum of irreducible subrepresentations U_j, and each is isomorphic to some $U \in \mathrm{Irr}(G)$ by an intertwiner $S : U \to U_j$. By Proposition 7.5(ii), S intertwines the action of a too, so

$$\rho_{U_j}(a) = S\rho_U(a)S^{-1} = 0.$$

Hence $\rho_V(a) = 0$ for every V. On the regular module, $\rho_{\mathrm{reg}}(a)$ is left multiplication by a, so

$$a = a \cdot e = \rho_{\mathrm{reg}}(a)\, e = 0.$$

Bijective. By Proposition 7.7,

$$\dim \bigoplus_U \mathrm{End}(U) = \sum_U (\dim U)^2 = \dim \mathbb{C}[G].$$

An injective linear map between spaces of the same dimension is bijective. ∎

Choosing bases identifies each block $\mathrm{End}(U)$ with $M_{\dim U}(\mathbb{C})$; $\Phi(a)$ has block $D_U(a)$.

For $C_2 = \{e, t\}$ and $a, b \in \mathbb{C}$, take the blocks in the order $\mathbf{1}, \mathrm{sgn}$. The element t acts by 1 and -1 on them, so the block map is

$$\Phi(ae + bt) = (a + b, a - b).$$

7.3 The centre and the class count

Definition 7.10 (Centre, class sums)**.** The *centre* of an algebra A is $Z(A) = \{z \in A : za = az \text{ for all } a \in A\}$, a commutative subalgebra. For a conjugacy class $\mathcal{K}$ of G, the *class sum* is $\widehat{\mathcal{K}} = \sum_{g \in \mathcal{K}} g \in \mathbb{C}[G]$.

The centre $Z(G)$ of the group and the centre $Z(\mathbb{C}[G])$ of its algebra are different objects: $Z(S_3) = \{e\}$, while $Z(\mathbb{C}[S_3])$ will have dimension 3. A class is written $\mathcal{K}$, not K, which Lecture 8 keeps for a subgroup.

Theorem 7.11 (The centre and the class count)**.**

(i) *The class sums are a basis of* $Z(\mathbb{C}[G])$.

(ii) Φ *maps* $Z(\mathbb{C}[G])$ *onto* $\bigoplus_U \mathbb{C}\,\mathbb{1}_U$*: each central* z *acts on each* $U \in \mathrm{Irr}(G)$ *as a scalar. Hence* G *has as many irreducible representations, up to isomorphism, as conjugacy classes, and the irreducible characters are an orthonormal basis of* $\mathrm{CF}(G)$.

(iii) (Column orthogonality.) *For* $g, h \in G$,

$$\sum_{U \in \mathrm{Irr}(G)} \overline{\chi_U(g)}\, \chi_U(h) = \begin{cases} |C_G(g)| & \text{if } h \in [g], \\ 0 & \text{otherwise.} \end{cases}$$

Proof. (i) As G spans $\mathbb{C}[G]$, a is central exactly when $ha = ah$, that is $hah^{-1} = a$, for every $h \in G$:

$$\sum_x a_x\, hxh^{-1} = \sum_x a_x\, x \qquad \text{for all } h \in G.$$

The coefficient of hxh^{-1} on the left is a_x, so this says $a_{hxh^{-1}} = a_x$ for all h and x: the coefficients are constant on each class. These a are the linear combinations of the class sums, which are linearly independent, as distinct classes are disjoint.

(ii) An isomorphism of algebras maps centre onto centre, and the centre of a direct sum is the direct sum of the centres. If T is in the centre of $\mathrm{End}(U)$, then T commutes with every $\rho_U(g)$, so $T \in \mathrm{End}_G(U) = \mathbb{C}\mathbb{1}$ by Schur's lemma (Theorem 5.13). Conversely, a scalar multiple of $\mathbb{1}_U$ commutes with every endomorphism. The class-sum basis therefore gives

$$\#\text{classes} = \dim Z(\mathbb{C}[G]) = \dim \bigoplus_U \mathbb{C}\,\mathbb{1}_U = |\mathrm{Irr}(G)|\,.$$

The irreducible characters are orthonormal (Corollary 6.8(i)), and there are $\dim \mathrm{CF}(G)$ of them.

(iii) Let $\mathcal{K}_1, \dots, \mathcal{K}_k$ be the classes, with $g_j \in \mathcal{K}_j$, and $\mathrm{Irr}(G) = \{U_1, \dots, U_k\}$, by the class count. Let M be the $k \times k$ matrix with entries

$$M_{ij} = \sqrt{|\mathcal{K}_j| / |G|}\; \chi_{U_i}(g_j).$$

In the class form of $\langle \cdot, \cdot \rangle_G$ (§ 6.1), the rows are orthonormal:

$$\overline{M} M^{\mathsf{T}} = \mathbb{1}.$$

A square matrix with a right inverse has it as a left inverse, so

$$M^{\mathsf{T}} \overline{M} = \mathbb{1},$$

and conjugating its entries,

$$\sum_i \overline{M_{ij}}\, M_{ij'} = \delta_{jj'}.$$

Multiply by $|G| / \sqrt{|\mathcal{K}_j|\,|\mathcal{K}_{j'}|}$ and use Theorem 1.20(i):

$$\begin{aligned} \sum_i \overline{\chi_{U_i}(g_j)}\, \chi_{U_i}(g_{j'}) &= \frac{|G|}{|\mathcal{K}_j|}\, \delta_{jj'} \\ &= \left|C_G(g_j)\right|\, \delta_{jj'}. \end{aligned}$$

Characters and centraliser sizes are constant on conjugacy classes, so this proves column orthogonality. ∎

For example, use the S_3 character table of Example 6.4. The columns for a transposition and a 3-cycle give

$$1 \cdot 1 + (-1) \cdot 1 + 0 \cdot (-1) = 0.$$

For the transposition column with itself, the sum is $1 + 1 + 0 = 2$, the order of its centraliser (Example 1.21).

The identity $\mathbb{1}_U$ in one block of Φ comes from a central element. Its name e_U, subscripted by an irreducible, is not the identity e of G.

Proposition 7.12 (Central idempotents). *For $U \in \mathrm{Irr}(G)$, let*

$$e_U = \frac{\dim U}{|G|} \sum_{g \in G} \overline{\chi_U(g)}\, g.$$

Then e_U is central, $\rho_U(e_U) = \mathbb{1}$, and $\rho_{U'}(e_U) = 0$ for $U' \in \mathrm{Irr}(G)$, $U' \neq U$. So $e_U^2 = e_U$, $e_U e_{U'} = 0$ for $U' \neq U$, and $\sum_U e_U = e$.

Proof. As Φ is bijective, exactly one $a \in \mathbb{C}[G]$ has $\Phi(a)$ equal to $\mathbb{1}_U$ in the block of U and 0 in the others. It is central, by Theorem 7.11(ii). We show $a = e_U$; the relations then follow, as Φ is injective.

Coefficients by traces. For $b \in \mathbb{C}[G]$,

$$\chi_{\mathrm{reg}}(b) = \sum_g b_g \chi_{\mathrm{reg}}(g) = |G|\ b_e.$$

The coefficient of e in $x^{-1}a$ is a_x, so

$$a_x = \frac{1}{|G|}\chi_{\mathrm{reg}}(x^{-1}a).$$

The coefficients of a. By Proposition 7.7, $\chi_{\mathrm{reg}} = \sum_{U'} \dim U'\, \chi_{U'}$, also on $\mathbb{C}[G]$, as both sides are linear. And $\rho_{U'}(x^{-1}a) = \rho_{U'}(x^{-1})\rho_{U'}(a)$ is $\rho_U(x^{-1})$ for $U' = U$, and 0 otherwise. So, by Proposition 6.2(ii),

$$\begin{aligned} a_x &= \frac{1}{|G|}\sum_{U'} \dim U'\, \chi_{U'}(x^{-1}a) \\ &= \frac{\dim U}{|G|}\chi_U(x^{-1}) = \frac{\dim U}{|G|}\overline{\chi_U(x)}. \end{aligned}$$ ∎

7.4 Commutants

Definition 7.13 (Commutant)**.** The *commutant* of a representation V of G is the algebra $\mathrm{End}_G(V)$ of its intertwiners $V \to V$.

Fix a representation V of G. For $U \in \mathrm{Irr}(G)$, write $M_U = \mathrm{Hom}_G(U, V)$ for the multiplicity space of U in V (Definition 5.15); its dimension is $m_U = m_U(V)$. The subscript of M_U is an irreducible representation; in the matrix algebra $M_n(\mathbb{C})$ the subscript is an integer. The group G acts on $U \otimes M_U$ through the first factor, $g \cdot (u \otimes f) = (g \cdot u) \otimes f$.

Proposition 7.14 (Isotypic components)**.**

(i) The map

$$\Theta\colon \bigoplus_{U \in \mathrm{Irr}(G)} U \otimes M_U \to V, \qquad u \otimes f \mapsto f(u),$$

is an isomorphism of representations.

(ii) Let $V_U = \Theta(U \otimes M_U)$, so that $V = \bigoplus_U V_U$. Then $\rho_V(e_U)$ is the projection of V onto V_U along the other components.

Proof. (i) *An intertwiner.* Evaluation $(u, f) \mapsto f(u)$ is bilinear, so it defines Θ on each tensor product. As f is an intertwiner,

$$\Theta\big((g \cdot u) \otimes f\big) = f(g \cdot u) = g \cdot f(u).$$

Onto. By Maschke's theorem, $V = U_1 \oplus \cdots \oplus U_k$ with each U_j irreducible. An isomorphism from some $U \in \mathrm{Irr}(G)$ onto U_j, viewed as a map into V, is an $f \in M_U$ with $\Theta(U \otimes f) = U_j$. So the image of Θ contains every U_j.

Bijective. Exactly m_U of the U_j are isomorphic to U (Corollary 5.16), so

$$\dim V = \sum_U m_U \dim U = \dim \bigoplus_U U \otimes M_U.$$

An onto linear map between spaces of the same dimension is bijective.

(ii) On $U' \otimes M_{U'}$, $a \in \mathbb{C}[G]$ acts as $\rho_{U'}(a) \otimes \mathbb{1}$. So, by Proposition 7.12, e_U acts as the identity on $U \otimes M_U$ and as 0 on the other summands. Θ intertwines the action of e_U (Proposition 7.5(ii)), so $\rho_V(e_U)$ is the identity on V_U and 0 on every other $V_{U'}$. ∎

Definition 7.15 (Isotypic component)**.** For $U \in \mathrm{Irr}(G)$, the subspace $V_U = \Theta(U \otimes M_U)$ is the *isotypic component* of U in V.

Example 7.16 (Guard: a component, not a copy)**.** Let E be the two-dimensional irreducible plane of S_3 from § 5.5, and let $V = E \oplus E$. Then $V = V_E$, so $\rho_V(e_E) = \mathbb{1}_V$, of rank 4, not 2. A central element acts on all of $V = V_E$ by one scalar (Theorem 7.11), so it cannot pick out just one copy of E. Moreover, for each $(c, c') \neq (0, 0)$, the intertwiner $u \mapsto (cu, c'u)$ has image a copy

$$E_{c,c'} = \{(cu, c'u) : u \in E\}.$$

If (c, c') and (d, d') are non-proportional, then $cd' - c'd \neq 0$, so $(cu, c'u) = (dw, d'w)$ forces $u = w = 0$. Their copies therefore intersect only at 0 and are different. Thus the copies themselves are not canonical. Lecture 14 needs this distinction.

Theorem 7.17 (The commutant)**.** *Let V be a representation of G. The map*

$$\Psi\colon \mathrm{End}_G(V) \to \bigoplus_{U \in \mathrm{Irr}(G)} \mathrm{End}(M_U), \qquad \Psi(T)_U(f) = T \circ f,$$

is an isomorphism of algebras. So $\dim \mathrm{End}_G(V) = \sum_U m_U^2$*, and* $\mathrm{End}_G(V)$ *is commutative if and only if every* $m_U \leq 1$.

Proof. A homomorphism. If $f\colon U \to V$ is an intertwiner, so is $T \circ f$; so $\Psi(T)_U \in \mathrm{End}(M_U)$. Composition gives $\Psi(ST) = \Psi(S)\Psi(T)$, and $\Psi(\mathbb{1})$ is the identity.

Injective. For $u \in U$ and $f \in M_U$,

$$T\,\Theta(u \otimes f) = Tf(u) = \Theta\big(u \otimes \Psi(T)_U f\big).$$

As Θ is onto, T is determined by $\Psi(T)$.

Onto. Given $B_U \in \mathrm{End}(M_U)$ for each U, let $T = \Theta \circ \big(\bigoplus_U \mathbb{1}_U \otimes B_U\big) \circ \Theta^{-1}$. It is an intertwiner, as Θ is one and each $\mathbb{1}_U \otimes B_U$ commutes with the action $\rho_U(g) \otimes \mathbb{1}$. For $u \in U$ and $f \in M_U$,

$$(Tf)(u) = \Theta(u \otimes B_U f) = (B_U f)(u),$$

so $\Psi(T)_U = B_U$.

The consequences. $\dim \mathrm{End}(M_U) = m_U^2$, and the dimension of the direct sum is the sum of these dimensions. The direct sum is commutative exactly when every block is. Now $\mathrm{End}(M)$ is commutative exactly when $\dim M \leq 1$: otherwise two matrix units have $E_{12}E_{21} \neq E_{21}E_{12}$. ∎

In matrices, $\mathrm{End}_G(V) \cong \bigoplus_U M_{m_U}(\mathbb{C})$; for V irreducible, this is Schur's lemma.

7.5 Centraliser algebras

For $H \leq G$, restricting ρ_U to H makes a representation U of G one of H, written $\operatorname{Res}^G_H U$ (Lecture 8 studies it).

Definition 7.18 (Multiplicity-free)**.** A representation of H is *multiplicity-free* if each irreducible representation of H occurs in it at most once.

Definition 7.19 (Centraliser algebras)**.** For $H \leq G$, the *centraliser algebra* is

$$\mathbb{C}[G]^H = \{a \in \mathbb{C}[G] : ha = ah \text{ for all } h \in H\}.$$

It is a subalgebra, and $\mathbb{C}[G]^G = Z(\mathbb{C}[G])$. The proof of Theorem 7.11(i), with $h \in H$, shows that $a \in \mathbb{C}[G]^H$ exactly when $a_{hxh^{-1}} = a_x$ for all $h \in H$ and $x \in G$. So the sums over the H-*classes* $\{hxh^{-1} : h \in H\}$ of G are a basis of $\mathbb{C}[G]^H$.

Theorem 7.20 (The Gelfand criterion)**.** *For $H \leq G$, the following are equivalent:*

(i) $\operatorname{Res}^G_H U$ *is multiplicity-free for every* $U \in \operatorname{Irr}(G)$*;*

(ii) $\mathbb{C}[G]^H$ *is commutative.*

Proof. The blocks of $\mathbb{C}[G]^H$*.* Let $a \in \mathbb{C}[G]$ and $h \in H$. As Φ is an injective homomorphism, $ha = ah$ exactly when $\Phi(h)\Phi(a) = \Phi(a)\Phi(h)$, that is, when each $\rho_U(a)$ commutes with $\rho_U(h)$. So Φ maps $\mathbb{C}[G]^H$ into $\bigoplus_U \operatorname{End}_H(\operatorname{Res}^G_H U)$. It maps onto it, since every element of the sum is $\Phi(a)$ for some a, which then lies in $\mathbb{C}[G]^H$. Hence

$$\mathbb{C}[G]^H \cong \bigoplus_{U \in \operatorname{Irr}(G)} \operatorname{End}_H\big(\operatorname{Res}^G_H U\big).$$

Commutativity. A direct sum of algebras is commutative exactly when each summand is. By Theorem 7.17, for the group H and the representation $\operatorname{Res}^G_H U$, the summand of U is commutative exactly when every irreducible representation of H occurs in $\operatorname{Res}^G_H U$ at most once. ∎

So $\dim \mathbb{C}[G]^H$ is the sum of the squared multiplicities in all the $\operatorname{Res}^G_H U$.

Example 7.21 (S_2 in S_3)**.** Let $S_2 = \{e, (1\,2)\} \leq S_3$, the stabiliser of 3. Conjugating by $(1\,2)$,

$$(1\,2)(1\,3)(1\,2)^{-1} = (2\,3),$$
$$(1\,2)(1\,2\,3)(1\,2)^{-1} = (1\,3\,2).$$

So there are four S_2-classes in S_3, and their sums are a basis of $\mathbb{C}[S_3]^{S_2}$, of dimension 4:

$$e, \quad (1\,2), \quad (1\,3)+(2\,3), \quad (1\,2\,3)+(1\,3\,2).$$

It is commutative. The last element is a class sum of S_3, and

$$(1\,3)+(2\,3) = \big[(1\,2)+(1\,3)+(2\,3)\big] - (1\,2),$$

with a class sum in brackets. So $\mathbb{C}[S_3]^{S_2}$ is spanned by $(1\,2)$ and elements central in $\mathbb{C}[S_3]$, which commute pairwise.

By Theorem 7.20, each irreducible representation of S_3 restricts to S_2 without multiplicity. The character table gives $\operatorname{Res}^{S_3}_{S_2} E \cong \mathbf{1} \oplus \operatorname{sgn}$, while $\mathbf{1}$ and sgn restrict to $\mathbf{1}$ and sgn; the squared multiplicities add up to the dimension,

$$1^2 + 1^2 + (1^2 + 1^2) = 4.$$

For $H = \{e\}$ both sides fail: $\mathbb{C}[S_3]$ is not commutative, and E restricts to $\mathbf{1} \oplus \mathbf{1}$. Lecture 10 applies the criterion to $S_{n-1} \le S_n$, and meets $(1\,3) + (2\,3)$ again, as a Jucys–Murphy element defined there.

7.6 Worked example

We compute the blocks of $\mathbb{C}[S_3]$ and the commutant of S_3 on $\mathbb{C}^3 \otimes \mathbb{C}^3$.

The blocks. $\operatorname{Irr}(S_3) = \{\mathbf{1}, \operatorname{sgn}, E\}$, of degrees 1, 1, 2 (Example 6.10). So, in Lecture 5's basis of E,

$$\mathbb{C}[S_3] \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}).$$

The idempotent of E. Substitute Example 6.4's character values:

$$e_E = \frac{1}{3}\big(2e - (1\,2\,3) - (1\,3\,2)\big).$$

Let $c = (1\,2\,3)$. The matrix of $D_E(c)$ in § 5.5 gives $D_E(c)^2 + D_E(c) = -\mathbb{1}$, and $D_E(c^{-1}) = D_E(c)^2$. Both 3-cycles act by 1 on $\mathbf{1}$ and sgn, where $\frac{1}{3}(2 - 1 - 1) = 0$; on E the formula gives $\frac{1}{3}(2\mathbb{1} + \mathbb{1}) = \mathbb{1}$. Thus $\Phi(e_E) = (0, 0, \mathbb{1})$, as Proposition 7.12 predicts.

On $\mathbb{C}^3$. $\mathbb{C}^3 = L \oplus E$ with $L \cong \mathbf{1}$ (§ 5.5). By Proposition 7.12, e_E acts as $\mathbb{1}$ on E and as 0 on $\mathbf{1}$, hence on L by Proposition 7.5(ii). So e_E projects onto E along L:

$$e_E \cdot x = x - \frac{1}{3}(x_1 + x_2 + x_3)(1, 1, 1).$$

A commutant. Let $V = \mathbb{C}^3 \otimes \mathbb{C}^3$, that is, $(\mathbf{1} \oplus E)^{\otimes 2}$. By Example 5.9 and Theorem 6.13(ii), $\chi_V(g) = |\operatorname{Fix}(g)|^2$. Using the character table gives

$$\begin{aligned} m_{\mathbf{1}} &= \frac{1}{6}(9 + 3 \cdot 1 + 2 \cdot 0) = 2, \\ m_{\operatorname{sgn}} &= \frac{1}{6}(9 - 3 \cdot 1 + 2 \cdot 0) = 1, \\ m_E &= \frac{1}{6}(2 \cdot 9 + 3 \cdot 0 \cdot 1 + 2 \cdot (-1) \cdot 0) = 3, \end{aligned}$$

and $V \cong \mathbf{1}^{\oplus 2} \oplus \operatorname{sgn} \oplus E^{\oplus 3}$. By Theorem 7.17,

$$\operatorname{End}_{S_3}(V) \cong M_2(\mathbb{C}) \oplus \mathbb{C} \oplus M_3(\mathbb{C}),$$

of dimension

$$4 + 1 + 9 = 14.$$

Self-check. What is the rank of e_E on V? *Answer:* $e_E^2 = e_E$ by Proposition 7.12, so $\rho_V(e_E)$ is an idempotent and its rank is its trace (Lemma 5.7):

$$\begin{aligned} \operatorname{tr} \rho_V(e_E) &= \frac{1}{3}\big(2\chi_V(e) - \chi_V((1\,2\,3)) - \chi_V((1\,3\,2))\big) \\ &= \frac{1}{3}(18 - 0 - 0) = 6. \end{aligned}$$

This is $\dim V_E = 3 \cdot 2$: the three copies of E together, not one (Example 7.16).

Summary.

- *The main results:* $\sum_U (\dim U)^2 = |G|$ and $\mathbb{C}[G] \cong \bigoplus_U \mathrm{End}(U)$ (Proposition 7.7 and Theorem 7.9); as many irreducibles as classes (Theorem 7.11); the central idempotents project onto the isotypic components (Proposition 7.14); and $\mathrm{End}_G(V) \cong \bigoplus_U \mathrm{End}(M_U)$ (Theorem 7.17).
- For $H \leq G$, every irreducible G-representation restricts multiplicity-free to H exactly when $\mathbb{C}[G]^H$ is commutative (Theorem 7.20).
- *Handed on:* $\mathbb{C}[G]$ and its modules to Lecture 8; class sums acting by scalars and column orthogonality to Lecture 9; the Gelfand criterion, with $\mathbb{C}[S_3]^{S_2}$, to Lecture 10; isotypic components and commutants to Lecture 14.
- *Next:* Lecture 8 constructs representations from subgroup representations.

References

Serre, *Linear Representations of Finite Groups*: the regular representation, Prop. 5 and its corollaries, p. 18; the number of irreducibles, Thms 6–7, pp. 19–20, and column orthogonality, Prop. 7, p. 20; the canonical decomposition and its projections, Thm 8, p. 21; the group algebra, §6.1, p. 47; $\mathbb{C}[G] \cong \prod_i \mathrm{End}(W_i)$, Prop. 10, p. 48; class sums and the centre, §6.3, p. 50.[1] Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: the commutant, Thm 1.2.14 and Cor. 1.2.17, pp. 16–17; the centraliser algebra and multiplicity-free restriction, Thm 2.1.10, p. 64.

[1] Serre's scalar product is antilinear in the second slot; his projection formula carries the conjugate where ours does.